Epidemic Simulation

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Introduction

Epidemic simulation research using a susceptible infected recovered model to recover how populations change in susceptible, infected, and removal when there is an epidemic in one or more places. The SIR model is often used in places where the population is small. For example, village, college community, and something else. The research will look at the changes in population as the number of people moves between two and three cities through graphs. There will be three methods for changing the foundational functions to show the situation. In the basic SIR model, the equations are as follows:

$$\frac{dS}{dt} = -\alpha \cdot S \cdot I,$$

$$\frac{dI}{dt} = \alpha \cdot S \cdot I - \beta \cdot I,$$

$$\frac{dR}{dt} = \beta \cdot I,$$

where α is the infected rate, β is the recovery rate,S represents susceptible individuals, I represents infected individuals, and R represents recovered individuals

Model 1

$$\frac{\Delta S}{dt} = -\alpha \cdot S \cdot I - U_{1-2} + T_{2-1}$$

$$\frac{\Delta I}{dt} = \alpha \cdot S \cdot I - \beta \cdot I - U_{1-2} + T_{2-1}$$

$$\frac{\Delta R}{dt} = \beta \cdot I - U_{1-2} + T_{2-1}$$

I propose a method to improve this prediction using the dollar sign **$** feature. This enhancement aims to provide a more precise understanding of population movements, thereby refining epidemic modeling and intervention strategies.

Change sizes of cities

When City 1 has 1000 people, and City 2 has 2000 people. So let $E2/1000 \times \$H\$2$ people from city 1 to city 2, $E2/1000$ be people who are not sick in the total population in the city, and H2 is the fixed value 5. It lets $I2/2000 \times \$L\$2$ people in city 2 go to city 1. The fixed number is 5 too.

Fig. 1: Flow between two cities

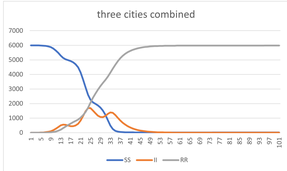

Fig. 2: Flow between three cities

Using the same functions in three cities. City 1 has 1000 people, the population of City 2 has 3000 people, and there are 2000 people in City 3 at the beginning. Assumed the epidemic broke out from City 1, and the fixed value changed from 5 to 1.

Results

With a daily flow of one person from City 1 to City 2 and one person from City 2 to City 3 in each category of susceptible, infected, and recovered individuals. After adjusting the population sizes of the cities, in the scenario where the initial infections occur in the smallest city, a notable trend emerges the smaller the city, the quicker the rise to peak infections. Furthermore, in larger cities, the total number of infected individuals tends to be higher.

Model 2

In real life, there will be a natural number of people flowing between cities. I let five people from each city flow between cities. So, setting one rand column and the other three columns. These columns use some functions to let every time choose one person in one city go to the next city from S, I, R (S is susceptible people, I is infected people, R is recovery people). If we let five people from one city to another city, there will be five-rand groups.

$$\frac{\Delta S}{dt} = -\alpha \cdot S \cdot I - U_{1-2} + T_{2-1}$$

$$\frac{\Delta I}{dt} = \alpha \cdot S \cdot I - \beta \cdot I - U_{1-2} + T_{2-1}$$

$$\frac{\Delta R}{dt} = \beta \cdot I - U_{1-2} + T_{2-1}$$

Two Cities

In this scenario, five individuals from each city traverse between cities daily. When the recovery rate is raised from 0.2 to 0.5, a decrease in the number of infected individuals is observed. With the augmented recovery rate, the decrement in infected individuals outpaces new infections, leading to a gradual increase in the infected count. Furthermore, altering the infected rate from 1/762 to 1/1000 results in a reduction in the infected population. As the infection rate diminishes, the rate at which new individuals are added to the infected group slows down. Lowering the infected rate further, such as to 1/1200, may lead to negative values in the infected population column. This occurs because the subtraction operation of the second function exceeds the addition operation, resulting in a net decrease in the infected count.

When letting only two people from each city flow and a small city has one people get sick at the beginning, there are no people sick in the big city. Therefore, there are no people who get sick in about 20 days in the big city. And City 1 has a small number of population get sick. People who are in City 2 get sick because of infected people from City 1 spreading.

Fig. 3: Population for City 1

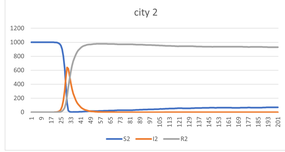

Fig. 4: Population for City 2

Three Cities

In three cities, in the same situation (city 1 has 300 people, city 2 has 700 people, city 3 has 1000 people) Comparing one person from each city flow with ten people from each city flow. There are similar numbers of infected people. The peak of infected people is similar too.

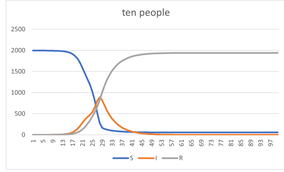

Fig. 5: One person flow

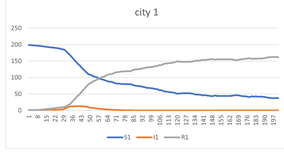

Fig. 6: Ten people flow

In three cities, whenever, one person from each city flows or ten people from each city flow if changing the infected rate to smaller or changing the recovery rate to bigger, City 2 and City 3 will have almost no person getting sick in a long time at the beginning.

Change population

I change the population of three cities. If City 1 has 500 people, City 2 has 1000 people, City 3 has 1200 people.

If there are the same number (10) of people from City 2 going City 3 as City 3 going City 2, City 3 will keep a balance, and 1200 people not sick and 0 people get sick. But one day there are two people from City 2 who get sick arrive in City 3. City 3 will have two people get sick.

Reasons

The reason S keeps negative is if S/1200<0.5, S<600, 0.5<(S+I)/1200, S+I>600, there will be one person from I go to City 2. Since there is one person who gets sick, minus so big number and the number of S becomes negative. In the beginning, time after there is one person who gets sick in City 3, City 3 fits the condition. After some days, since S is negative,1/762*S*I will become smaller, and S+I will become smaller. S+I<600, S+I/1200<0.5, so, there will be one person from R going to City 2 and from I and R more people are going to City 2. And from S, in City 3 more people are coming and fewer people going, and the negative number will become positive.